

Yuriy Dementiev

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🌐 yudementiev.github.io

Positions

- 2024 – ····
- ITMO University
Head of TCS Lab
 - Computer Science Space
Co-Founder, Coordinator
 - A. A. Markov Lab for Computer Science, Mathematics and Artificial Intelligence, St. Petersburg University
Researcher
 - LLC “VMIRE”
Head of Analytics and Data Science
- 2022 – ····
- Leonhard Euler International Mathematical Institute
Researcher
- 2022 – 2024
- International Laboratory of Game Theory and Decision Making, HSE University
Researcher
 - MY.GAMES Development (VK)
Middle Data Analyst

Education

- 2024 – 2028
- Ph.D. Student in Theoretical Computer Science in PDMI RAS
In progress
Supervisor: Sergey Nikolenko
- 2022 – 2024
- M.Sc. in Mathematics and Computer Science in St. Petersburg University
Emphasys on Software Development and Data Science
Graduated cum laude
Thesis title: *Inconsistent Planning: Probabilistic Model*
Supervisor: Fedor Fomin
- 2018 – 2022
- B.Sc. in Mathematics and Computer Science in St. Petersburg University
Emphasys on Mathematics and Theoretical Computer Science
Graduated cum laude
Thesis title: *Half-duplex communication complexity bounds, and topological embedding problem for colored trees*
Supervisor: Alexander Smal

Research Interests

Algorithms, Parameterized Complexity, Algorithmic Game Theory, Communication Complexity, Data Analysis

Publications

Conference Proceedings

- 1 T. Belova, Y. Dementiev, F. Fomin, P. Golovach, and A. Ignatiev, “How to guide a present-biased agent through prescribed tasks?” In *ECAI 2024 - 27th European Conference on Artificial Intelligence, 19-24 October 2024, Santiago de Compostela, Spain - Including 13th Conference on Prestigious Applications of Intelligent Systems (PAIS 2024)*, 2024.  URL: <https://ebooks.iospress.nl/doi/10.3233/FAIA240898>.
- 2 N. Morozov, A. Ignatiev, and Y. Dementiev, “Several stories about high-multiplicity efx allocation,” in *AAAI Student Abstract*, 2024.
- 3 Y. Dementiev, F. Fomin, and A. Ignatiev, “Inconsistent planning: When in doubt, toss a coin!” In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022, pp. 9724–9731.  URL: <https://doi.org/10.1609/aaai.v36i9.21207>.
- 4 Y. Dementiev, A. Ignatiev, V. Sidelnik, A. Smal, and M. Ushakov, “New bounds on the half-duplex communication complexity,” in *Bure’s T. et al. (eds) SOFSEM 2021: Theory and Practice of Computer Science. Lecture Notes in Computer Science*, vol 12607. Springer, Cham, 2021.  URL: https://doi.org/10.1007/978-3-030-67731-2_17.

Preprints

- 1 Y. Dementiev, N. Gravin, and A. Ignatiev, *Integrality gap of nash welfare maximization with money*, 2025.
- 2 Y. Dementiev, F. Fomin, and A. Ignatiev, *Inconsistent planning: When in doubt, toss a coin!* 2022.  URL: <https://arxiv.org/abs/2112.03329>.
- 3 Y. Dementiev, A. Ignatiev, V. Sidelnik, A. Smal, and M. Ushakov, *New bounds on the half-duplex communication complexity*, 2021.  URL: <https://eccc.weizmann.ac.il/report/2020/117/>.

Theses

- 1 Y. Dementiev, “Inconsistent planning: Probabilistic model,” MSc thesis, Faculty of Math and Computer Science, St Petersburg University, 2024.
- 2 Y. Dementiev, “Half-duplex communication complexity bounds, and topological embedding problem for colored trees,” BSc thesis, Faculty of Math and Computer Science, St Petersburg University, 2022.  URL: <https://dspace.spbu.ru/handle/11701/40003>.

Teaching

Spring 2026	 Algorithms for NP-hard problems CS Space
Spring 2025	 Algorithms for NP-hard problems ITMO University
	 Discrete Algorithms of Fair Division Sirius
Spring 2024	 Information Theory St. Petersburg University
	 Communication Games Sirius
Summer 2023	 Inconsistent Planning Summer School in St. Petersburg University
Spring 2023	 Mathematical Foundations of Algorithms St. Petersburg University

Teaching (continued)

- 2022 - 2023  **Greedy Conjecture in the Superstring Problem**
Sirius
- 2022 - 2023  **Computer Science Seminar**
St. Petersburg University
- Summer 2022  **Supervisor of the research project for students in TCS**
St. Petersburg University
- Summer 2022  **Inconsistent Planning**
Summer School in St. Petersburg University
- 2020 - 2022  Math teacher in mathematical club for talented schoolchildren

Research Visits

- 2023  **Institute for Theoretical Computer Science, SUFE**
Shanghai, Host: Nick Gravin

Miscellaneous Experience

Talks and Posters

- 2026  **AAAI Conference**
Talk “Structural Approach to Guiding a Present-Biased Agent”
Talk “EFX and PO Allocation Exists for Two Types of Goods”
Singapore
-  **IJCAI Conference**
Talk “Subexponential Algorithm for High Multiplicity Fair Division”
Bremen, Germany
- 2025  **ECAI Conference**
Talk “Integrality Gap of Nash Welfare Maximization with Money”
Bologna, Italy
- 2024  **ECAI Conference**
Talk “How To Guide a Present-Biased Agent Through Prescribed Tasks?”
Santiago de Compostela, Spain
- 2022  **36th AAAI Conference**
Talk “Inconsistent Planning: When in doubt, toss a coin!”
Virtual Conference
-  **36th AAAI Conference**
Poster Session, online
- 2021  **SOFSEM Conference**
Conference, online

Student Schools

- 2023  **EPIT 2023: Le Kaléidoscope de la Complexité**
Oleron Island, France
- 2021  **Computational Complexity Theory**
Sirius University, Sochi

Miscellaneous Experience (continued)

Awards and Achievements

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| 2023 - 2024 | ■ | Winner of “ Competition for the support of young researchers in St. Petersburg ”
from Leonhard Euler International Mathematical Institute and St. Petersburg University |
| 2022 - 2023 | ■ | VK Scholarship |
| | ■ | Winner of “ Open competition to support young mathematicians of St. Petersburg ”
from Leonhard Euler International Mathematical Institute and St. Petersburg University |
| 2021 - 2022 | ■ | Winner of “ Competition of mathematical projects completed by young re-
searchers ”
from Leonhard Euler International Mathematical Institute |
| 2019 - 2022 | ■ | Gazprom Neft Scholarship |
| | ■ | Prizewinner of several olympiads in Mathematics and Informatics |